

PART-1 : MATHEMATICS**SECTION-I (i)**

1) Let S be the set of all polynomials $P(x)$ of degree 2026 with real coefficients such that $P(x) \geq 0$ for all $x \in \mathbb{R}$. For any $P \in S$, defined the auxiliary polynomial $Q(x)$ as the sum of $P(x)$ and all its derivatives :

$$Q(x) = P(x) + P'(x) + P''(x) + \dots + P^{(2026)}(x)$$

Which of the following statements **MUST** be true for the function $Q(x)$?

- (A) All the roots are real
- (B) There exists no real root
- (C) There exists atleast two real roots
- (D) There exists 1013 distinct real roots

2) Let $f(x) = x^3 - 3x + c$, where c is a real parameter. Consider the nested equation $f(f(x)) = 0$. Let N be the number of itneger values of c for which the equation $f(f(x)) = 0$ has exactly 9 distinct real roots. The value of N is :

- (A) 0
- (B) 1
- (C) 2
- (D) 3

3) Consider the function $f(x) = \frac{x^x}{(1-x)^{1-x}}$ defined on the open interval $(0, 1)$. Let M and m denote the local maximum and local minimum values of $f(x)$ respectively. The product $M \cdot m$ equals :

- (A) 1
- (B) e^{-2}
- (C) e
- (D) $1/e$

4) Let C be the curve defined by $y = \frac{x^4}{4} + \frac{x^2}{2}$. For every real number m , there exists a unique line L_m : $y = mx - c(m)$ which is tangent to the curve C . If the minimum value of the function $h(m) = c(m) - 2m$ is attained at $m = m_0$, then m_0 is equal to :

- (A) 2
- (B) 6
- (C) 10
- (D) 14

SECTION-I (ii)

1)

Let $f : [0,1] \rightarrow \mathbb{R}$ be a twice continuously differentiable function such that $f(0) = 0$, $f(1) = 0$ and $f''(x) < 0$ for $x \in (0,1)$. Which of the following statements is/are CORRECT ?

- (A) The function $g(x) = \frac{f(x)}{x}$ is strictly decreasing on $(0,1]$.
- (B) The function $h(x) = \frac{f(x)}{1-x}$ is strictly increasing on $[0,1)$.
- (C) There exists $c_1 \in (0,1)$ such that $f'(c_1) = \frac{f(c_1)}{1-c_1}$.
- (D) There exists $c_2 \in (0,1)$ such that $f'(c_2) = \frac{f(c_2)}{c_2}$.

2) Consider the problem of determining the number of distinct tangents that can be drawn from the origin $(0,0)$ to various curves $y = f(x)$. Which of the following statements is/are CORRECT ?

- (A) Exactly one tangent can be drawn from the origin to the curve $f(x) = e^x$ on $\mathbb{R}$.
- (B) Exactly two distinct tangents can be drawn from the origin to the curve $f(x) = x^4 - 2x^2 + 2$ on $\mathbb{R}$.
- (C) Exactly four distinct tangents can be drawn from the origin to the curve $f(x) = 12x^2 - x^4$ on $\mathbb{R}$.
- (D) No tangent can be drawn from the origin to the curve $f(x) = x \lfloor nx \rfloor$ on $(0, \infty)$.

3) Consider the algebraic function $f(x) = \frac{x^2 + 2}{\sqrt{x^2 + 1}}$ defined for all $x \in \mathbb{R}$, and the geometric function $A(t)$ representing the area of the triangle bounded by the coordinate axes and the tangent line to the curve $xy = 2$ at a point $\left(t, \frac{2}{t}\right)$ in the first quadrant. Which of the following statements is/are CORRECT ?

- (A) The absolute minimum value of $f(x)$ on $\mathbb{R}$ is 2.
- (B) The function $f(x)$ attains its absolute minimum at exactly two distinct real values of x .
- (C) The area $A(t)$ is a constant value, completely independent of the parameter t .
- (D) The absolute minimum value of $A(t)$ for $t > 0$ is exactly 2.

4) Consider the family of polynomial function $f_k(x) = x^4 + kx^3 + x^2$ parameterized by a real number k .

Which of the following statement regarding the critical points and extrema of $f_k(x)$ is/are CORRECT ?

- (A) For $k = 2$, the function has exactly two local minima and one local maximum.
- (B) For $k = \frac{4\sqrt{2}}{3}$, the function has exactly two distinct local extrema.
- (C) For $k = 1$, the function possesses a stationary point of inflection.
- (D) Irrespective of the value of k , the function can never have more than one local maximum.

5) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function satisfying the equation :

$$f(x) + \int_0^x (t-x)f(t) dt = e^{-x}$$

Which of the following properties describe the derivative and extrema of $f(x)$?

- (A) $f(x)$ has exactly one local extrema on the interval $(0, \infty)$
- (B) The global maximum value of $f(x)$ on $\mathbb{R}$ is strictly greater than 1.
- (C) $f'(x) > 0$ for all $x \in (3, 5)$
- (D) The global maximum of $f(x)$ occurs exactly at $x = 0$.

6) For each integer $n \geq 2$, define the polynomial $P_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$. Which of the following statements regarding the real roots and extrema of $P_n(x)$ is/are **CORRECT** ?

- (A) If n is even, the equation $P_n(x) = 0$ has no real roots.
- (B) If n is odd, the equation $P_n(x) = 0$ has exactly one real root.
- (C) The function $g_n(x) = P_n(x)e^{-x}$ has exactly one local extremum for all even n .
- (D) For any $n \geq 2$, the polynomials $P_n(x)$ and $P_{n-1}(x)$ do not share any common real root.

SECTION-II (i)

Common Content for Question No. 1 to 2

Consider the cubic polynomial curve $C: y = x^3 - 6x^2 + 5x + 1$. A sequence of real number $\{x_n\}$ is generated by a geometric iterative process on this curve. An initial point $x_0 = 1$ is chosen on the x -axis and its corresponding point on the curve is $P_0(x_0, y_0)$. For any integer $n \geq 0$, a tangent line is drawn to the curve C at the point $P_n(x_n, y_n)$. This tangent line intersects the cubic curve C again at a unique point P_{n+1} , whose x -coordinate is defined as x_{n+1} .

- 1) Determine the value of the term x_3 in this sequence.
- 2) Find the value of the term x_5 in this sequence.

Common Content for Question No. 3 to 4

In statistical optimization, the weighted sum of absolute deviations is a function used to find a point of central tendency that minimizes total transport cost or distance. For a set of points $a_1 < a_2 < \dots < a_n$ with corresponding positive weight $w_1, w_2, \dots, w_n$, the objective function is

defined as $D(x) = \sum_{k=1}^n w_k |x - a_k|$. The absolute minimum of this function is attained at the weighted median of the data set.

Consider a specific objective function defined for $x \in \mathbb{R}$:

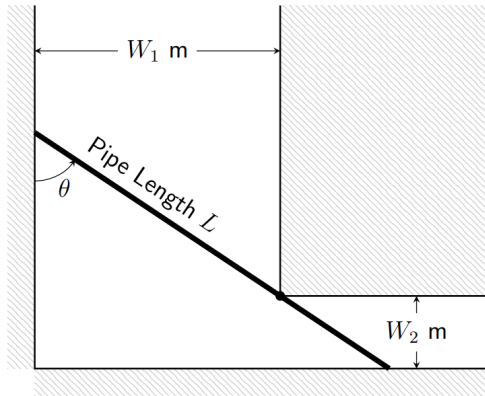
$$f(x) = 1|x - 1| + 2|x - 2| + 3|x - 3| + \dots + 10|x - 10|.$$

- 3) Let x_0 be the integer value at which the function $f(x)$ attains its absolute minimum. Find the value of x_0 .

4) Determine the absolute minimum value of the function $f(x_0)$.

Common Content for Question No. 5 to 6

A classic optimization problem involves maneuvering a rigid object through a restricted spatial corridor. Consider the top-down diagram below depicting two perfectly straight, perpendicular hallways that intersect at a right angled corner.



A rigid, ultra-thin pipe is to be carried horizontally around this corner. Let θ be the angle the pipe makes with the outer wall of the hallway of width W_1 when the pipe is jammed against the inner corner during the maneuver. The maximum length L of the pipe that can successfully negotiate the

turn is dictated by minimizing the geometric clearance function $L(\theta) = \frac{W_1}{\sin \theta} + \frac{W_2}{\cos \theta}$.

Assume the specific hallway widths are $W_1 = 27$ meters and $W_2 = 8$ meters. Let θ_c denote the critical angle that minimizes the clearance function.

5) Find the exact numerical value of $27 \cot^3 \theta_c$.

6) If $L_{\max}$ is the maximum possible length of the pipe that can be carried around this corner, find the value of $\frac{L_{\max}^2}{13}$

SECTION-II (ii)

1) For a specific positive real parameter c , the exponential curve $y = c^x$ and the power curve $y = x^5$ touch each other at exactly one point in the first quadrant, sharing a common tangent. Find the exact value of the expression $(e \ln c)$.

2) Let $f(x) = \frac{x^x e^x}{(x+1)^{x+1}}$ for $x > 0$. Let M be the absolute minimum value attained by $f(x)$ on its domain. Determine the value of $M + \frac{1}{e}$

3) Determine the smallest non-negative real value of k such that the inequality $x^4 - 4x^3 + kx^2 - 4x + 1 \geq 0$ holds true for all $x \in \mathbb{R}$.

PART-2 : PHYSICS

SECTION-I (i)

1) A hollow conducting spherical shell has inner radius R and outer radius $2R$. A point charge $+Q$ is fixed at a point inside the cavity, not at the centre. The shell is initially neutral and electrically isolated. A second conductor, completely enclosing the first shell but not touching it, is connected to earth. Electrostatic equilibrium is established. Which one of the following statements is correct?

- (A) The electric field inside the conducting material of the inner shell is zero, and the net charge on its outer surface is zero.
- (B) The electric field inside the conducting material of the inner shell is non-zero because the charge is placed asymmetrically.
- (C) The inner surface of the shell acquires a total charge $-Q$, the outer surface acquires a total charge $+Q$, irrespective of grounding of the outer conductor.
- (D) The total charge induced on the inner surface of the shell depends on the distance of the point charge from the centre.

2) A cube of side a has one vertex at $O(0, 0, 0)$, where a particle of charge $+q$ and mass m is placed.

Position	Charge
$(a, 0, 0)$	$+q$
$(0, a, 0)$	$+2q$
$(0, 0, a)$	$+3q$
$(a, a, 0)$	$-2\sqrt{2}q$
$(a, 0, a)$	$-4\sqrt{2}q$
$(0, a, a)$	$-6\sqrt{2}q$
(a, a, a)	$+3\sqrt{3}q$

Charges fixed at the remaining seven vertices are: All charges except the particle at O are fixed. The particle at O is released from rest. Neglect gravity. Its initial acceleration is:

- (A) $\frac{\sqrt{3}kq^2}{ma^2}$, along the body diagonal from O towards (a, a, a)
- (B) $\frac{3kq^2}{ma^2}$, along the body diagonal from O towards (a, a, a)
- (C) $\frac{\sqrt{3}kq^2}{ma^2}$, along the body diagonal directed away from (a, a, a)
- (D) $\frac{kq^2}{ma^2}$, along the direction $\frac{\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}}{\sqrt{14}}$.

3) A satellite moves in a circular orbit of radius $3R$ around a spherical planet of mass m and radius R . At a point P , an instantaneous impulse changes its speed to $\sqrt{\frac{3}{2}}$ times its original circular speed

and directs its velocity at 60° to the local tangent, towards the planet. Neglect atmosphere and planetary rotation. Which statement is correct ?

- (A) The satellite follows an ellipse of semi-major axis $6R$, does not collide with the planet, and has perigee distance $\frac{3R}{2} (4 - \sqrt{13})$.
- (B) The satellite follows an ellipse of eccentricity $\frac{\sqrt{13}}{4}$ and strikes the planet with its velocity making an angle $\tan^{-1} \sqrt{\frac{17}{27}}$ with the local tangent.
- (C) The satellite follows a hyperbolic trajectory and strikes the planet with speed $\sqrt{\frac{11GM}{6R}}$.
- (D) The satellite follows an ellipse of eccentricity $\frac{3}{4}$ and just grazes the planet's surface.

4) Two infinitely long, parallel line charges carrying uniform linear charge densities $+\lambda$ and $-\lambda$ are placed along the lines $x = -a, y = 0$ and $x = +a, y = 0$, respectively, both parallel to the z -axis. The potential is chosen to be zero on the plane $x = 0$. Which of the following represents the equipotential surface on which

$V = \frac{\lambda}{2\pi\epsilon_0} \ln 2$? ($\lambda > 0, a > 0$)

- (A) $\left(x - \frac{5a}{3}\right)^2 + y^2 = \left(\frac{4a}{3}\right)^2$
- (B) $\left(x + \frac{5a}{3}\right)^2 + y^2 = \left(\frac{4a}{3}\right)^2$
- (C) $\left(x - \frac{a}{3}\right)^2 + y^2 = \left(\frac{4a}{3}\right)^2$
- (D) $x^2 + y^2 = \left(\frac{3a}{2}\right)^2$

SECTION-I (ii)

1) Each option describes an independent charge distribution in vacuum. Take $k = \frac{1}{4\pi\epsilon_0}$. Which of the following statements are correct ? (λ_0, σ_0 are constants).

- (A) A thin rod occupies $-a \leq x \leq a$ on the x -axis and has linear charge density $\lambda(x) = \lambda_0 \frac{|x|}{a}$. The electric field at $P(0, a)$ is $\vec{E}_P = \frac{k\lambda_0}{a} \left(1 - \frac{1}{\sqrt{2}}\right) \hat{j}$.
- (B) A circular ring of radius a , lying in the xy -plane, has linear charge density $\lambda(\phi) = \lambda_0 \cos \phi$, where ϕ is measured from the positive x -axis. The electric field at its centre is $\vec{E} = -\frac{\pi k \lambda_0}{a} \hat{j}$.

- (C) A circular disc of radius a , lying in the xy -plane, has surface charge density $\sigma(r) = \sigma_0 \frac{r}{a}$. The electric field at $(0, 0, a)$ is $\vec{E} = 2\pi k \sigma_0 \left[\ln(1 + \sqrt{2}) - \frac{1}{\sqrt{2}} \right] \hat{k}$.
- (D) A hemispherical shell of radius a , occupying $z \geq 0$, carries uniform surface charge density σ . The electric field at its centre is $\vec{E} = -2\pi k \sigma \hat{k}$.

2) Two infinite parallel conducting plates A and B are located at $x = 0$ and $x = 3d$, respectively. Their potentials are maintained such that $V_B - V_A = -\frac{3\sigma d}{\epsilon_0}$. Two infinite non-conducting sheets are placed parallel to the plates :

- at $x = d$, with surface charge density $+\sigma$;
- at $x = 2d$, with surface charge density $-\sigma$.

The region between all sheets is vacuum. Edge effects are neglected, and $\sigma > 0$. Which statements are correct ?

- The electric fields in the three regions $0 < x < d$, $d < x < 2d$, and $2d < x < 3d$ are respectively
- (A) $\frac{\sigma}{\epsilon_0} \hat{i}, \frac{2\sigma}{\epsilon_0} \hat{i}, \vec{0}$.
- (B) The charge densities induced on the facing surfaces of plates A and B are respectively $+\sigma$ and 0 .
- (C) The potential of the sheet at $x = 2d$, relative to plate A, is $V(2d) - V_A = -\frac{3\sigma d}{\epsilon_0}$.
- (D) If the sheet at $x = d$ is shifted quasistatically to $x = \frac{d}{2}$, while all charge densities and plate potentials remain unchanged, the field in the region immediately to its right becomes $\frac{11\sigma}{6\epsilon_0} \hat{i}$.

3) Each option refers to an independent charge distribution in vacuum. All conducting surfaces are in electrostatic equilibrium. Which of the following statements are correct ?

- An isolated conducting sphere of radius R carries charge Q . The outward electrostatic pressure
- (A) on its surface is $p = \frac{Q^2}{16\pi^2 \epsilon_0 R^4}$.
- A spherical capacitor consists of concentric conducting shells of radii a and b ($b > a$). The inner and outer conductors carry charges $+Q$ and $-Q$, respectively. The electrostatic pressure on the
- (B) outer surface of the inner conductor is $p = \frac{Q^2}{32\pi^2 \epsilon_0 a^4}$, directed radially outward.
- A very long conducting cylindrical shell of radius R carries charge λ per unit length. The
- (C) outward electrostatic pressure on its curved surface is $p = \frac{\lambda^2}{8\pi^2 \epsilon_0 R^2}$.
- An infinite non-conducting sheet carries a uniform surface charge density σ . The net normal electrostatic force per unit area on an infinitesimal element of the sheet due to all the
- (D) remaining charges on the sheet is $p = \frac{\sigma^2}{(2\epsilon_0)}$.

4) A distant exoplanet is modelled as a non-rotating, uniform solid sphere of mass M and radius R . A

straight, frictionless tunnel is drilled along a chord whose minimum distance from the planet's

centre is $\frac{R}{2}$. A probe is initially held at rest at point P on the extension of the tunnel, where its distance from the planet's centre is $2R$. It is released and constrained to move along the tunnel and its straight-line extension. The tunnel walls exert no force along the direction of motion. Neglect

atmosphere and all external gravitational fields. Let $\omega = \sqrt{\frac{GM}{R^3}}$. Which of the following statements are correct ?

- (A) The speed of the probe just as it enters the planet is $\sqrt{\frac{GM}{R}}$.
The maximum speed of the probe is attained at the point of closest approach to the centre and
- (B) is $\sqrt{\frac{7GM}{4R}}$.
- (C) The time spent by the probe inside the planet during its first passage is $2\sqrt{\frac{R^3}{GM}} \sin^{-1} \sqrt{\frac{3}{7}}$.
- (D) The kinetic energy of the probe at the point of closest approach is $\frac{7}{2}$ times the kinetic energy of an identical satellite moving in a circular orbit just above the planet's surface.

5) A non-rotating astronomical body is modelled as a function solid sphere of mass M and radius R , centred at the origin $O(0, 0, 0)$. Two identical black holes, each of mass m , are fixed on the x -axis at $x = +D$ and $x = -D$, symmetrically about the sphere.

Given

$$D \gg R, \quad D \gg \frac{2Gm}{c^2},$$

Newtonian gravity is applicable throughout the sphere. Assume that the sphere remains undeformed with uniform mass density, and take the gravitational potential to be zero at infinity. Define

$$\lambda = \frac{2mR^3}{MD^3}.$$

Considering terms only up to the lowest non-vanishing order in $\frac{R}{D}$, which of the following statements are correct ?

- (A) The total gravitational potential at the centre of the sphere is $V_0 = -\frac{3GM}{2R} - \frac{2Gm}{D}$.
For a point (x, y, z) sufficiently close to the centre,
- (B) $V - V_0 = \frac{GM}{2R^3} \left[(1 - 2\lambda)x^2 + (1 + \lambda)(y^2 + z^2) \right]$.
- (C) The centre of the sphere is a stable equilibrium position for a test particle if $\lambda < \frac{1}{2}$.
- (D) If $\lambda > \frac{1}{2}$, the equilibrium at the centre is unstable for small displacements along the x -axis but remains stable for sufficiently small displacements confined to the yz -plane.

6) A binary star system consists of two stars of masses M and $2M$, revolving in circular orbits about their common centre of mass under mutual gravitational attraction. The distance between the stars

remains constant and equal to a . The stars may be treated as point masses, and the effect of all other astronomical bodies may be neglected. Which of the following statements are correct ?

- (A) The common angular speed of the two stars is $\omega = \sqrt{\frac{3GM}{a^3}}$.
- (B) The star of mass M moves in a circle of radius $\frac{2a}{3}$ about the centre of mass, and its orbital speed is $v = \sqrt{\frac{4GM}{3a}}$.
- (C) The total mechanical energy of the binary system in the centre-of-mass frame is $E = -\frac{GM^2}{a}$.
If L_M and L_{2M} are the magnitude of angular momenta of the stars of masses M and $2M$, respectively, about the common centre of mass, then $L_M = 2L_{2M}$, and the total angular
- (D) momentum of the system is $L = \frac{2M}{3}\sqrt{3GMa}$.

SECTION-II (i)

Common Content for Question No. 1 to 2

A spacecraft of negligible mass moves under the gravitational attraction of Earth, which may be treated as a point mass M . Atmospheric resistance and the gravitational influence of other celestial bodies are neglected. At a point P , the spacecraft is at a distance r_0 from Earth's centre. Define

$\nu_c = \sqrt{\frac{GM}{r_0}}$, where ν_c is the speed required for a circular orbit of radius r_0 . The velocity of the spacecraft at P has a radially outward component ν_r , and a transverse component ν_t .

1) The spacecraft is projected from P with $\nu_r = \frac{\nu_c}{2}$, $\nu_t = \nu_c$. If a is the semi-major axis of its orbit, calculate $\frac{a}{r_0}$.

2) In another launch from P , the velocity components are $\nu_r = \nu_c$, $\nu_t = \sqrt{2}\nu_c$. If r_p denotes the periaapsis distance of the resulting trajectory, calculate $\frac{r_p}{r_0}$.

Common Content for Question No. 3 to 4

In a two-dimensional region, two electrostatic fields are given by $\vec{E}_1 = E_0 (y\hat{i} + x\hat{j})$ and $\vec{E}_2 = \frac{\alpha}{x^2 + y^2} (x\hat{i} + y\hat{j})$, $x^2 + y^2 > 0$. The electric potential V is related to the electric field through $\vec{E} = -\nabla V$. For the field $\vec{E}_1$, take $V = 0$ at the origin. For the field $\vec{E}_2$, take $V = 0$ on the circle $r = 1$ m, where $r = \sqrt{x^2 + y^2}$. Use $E_0 = 3.0 \text{ Vm}^{-2}$, $\alpha = 4.0 \text{ V}$, $\ln 5 = 1.609$.

3) For the field $\vec{E}_1$, find the magnitude of electric potential, in volts, at the point $P = (2\text{m}, 1\text{m})$.

4) For the field $\vec{E}_2$, find the magnitude of electric potential, in volts, at the point $Q = (3m, 4m)$.

Common Content for Question No. 5 to 6

Four independent charge distributions, each carrying the same total charge Q , are considered. The

electric potential is zero at infinity and $k = \frac{1}{4\pi\epsilon_0}$. The distributions are :

(1) A solid sphere of radius R with $\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right)$, $0 \leq r \leq R$.

(2) A thick spherical shell occupying $R < r < 2R$ with $\rho(r) = \frac{\alpha}{r}$.

(3) A thin circular disc of radius R with $\sigma(r) = \sigma_0 \frac{r}{R}$, $0 \leq r \leq R$.

(4) A uniformly charged thin rod of length $2R$, lying along the x -axis from $x = -R$ to $x = R$.

Let V_1, V_2, V_3 be the potentials at the centres of the first three distributions, respectively, and let V_4

be the potential due to the rod at $\left(0, \frac{3R}{4}\right)$. Use $\ln 3 = 1.0986$.

5) Calculate $\frac{R(V_1 + V_2)}{kQ}$. Give your answer up to two decimal places.

6) Calculate $\frac{R(V_3 + V_4)}{kQ}$. Give your answer up to two decimal places.

SECTION-II (ii)

1) A non-conducting solid sphere of radius R carries a spherically symmetric volume charge density $\rho(r) = \rho_0 \frac{r}{R}$, $0 \leq r \leq R$, such that its total charge is $+Q$. It is surrounded by a concentric, infinitesimally thin insulating spherical shell of radius $2R$ carrying charge $-Q$, uniformly distributed over its surface. The complete charge configuration is assembled quasistatically from infinity. If its

total electrostatic potential energy is written as $U = \frac{n}{28} \left(\frac{1}{4\pi\epsilon_0} \right) \frac{Q^2}{R}$, find n .

2) An ideal electric dipole of dipole moment $\vec{p} = p\hat{i}$ is placed at the origin. A point P has coordinates $P \equiv (3a, 4a)$. A point charge Q , whose sign is not specified, is fixed at $A \equiv (3a, 3a)$. The resultant electric field at P , due to the dipole and the point charge, is directed parallel to the vector $\hat{i} + \hat{j}$. If $-\frac{3125Qa}{p} = n$, find the integer n .

3) A spherical planet of uniform density, mass M , and radius R rotates uniformly about its diameter

with angular speed ω . Define $g_s = \frac{GM}{R^2}$, $\omega^2 R = \frac{g_s}{2}$. A small test body is held stationary relative to

the planet at an interior point P , whose distance from the planet's centre is $r_P = \frac{3R}{4}$. The latitude of

P is λ , where $\cos \lambda = \frac{2}{3}$. Let g_p be the magnitude of the effective acceleration at P , including

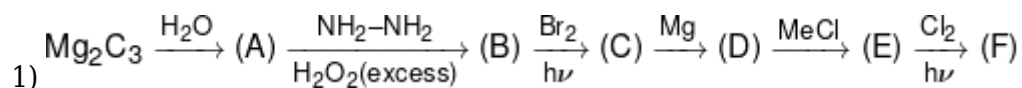
gravitational and centrifugal effects.

Another test body is held stationary at point Q, situated on the planet's rotation axis at a distance $2R$ from its centre. Let g_Q be the magnitude of its effective acceleration.

Neglect the gravitational influence of all other astronomical bodies. If $\left(\frac{g_P}{g_Q}\right)^2 = n$, find the integer n .

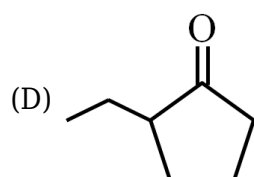
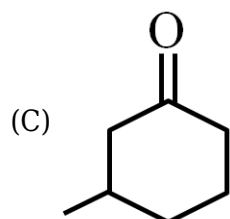
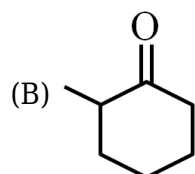
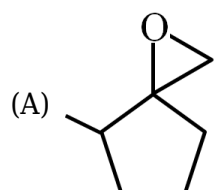
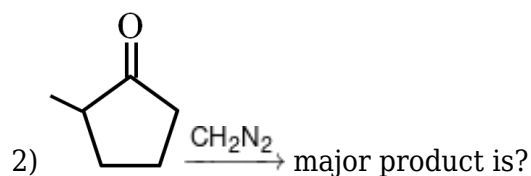
PART-3 : CHEMISTRY

SECTION-I (i)



A to F are major organic products. Select the correct statement about (F) ?

- (A) It can also be formed on reaction of isobutylene with $\text{HCl}/\text{R}_2\text{O}_2$.
- (B) With $\text{Na}/\text{Et}_2\text{O}$ it gives major product having 4 carbons.
- (C) With aq. KOH it gives an alcohol as major product.
- (D) With alc. KOH it gives two alkenes



3) Select correct reaction

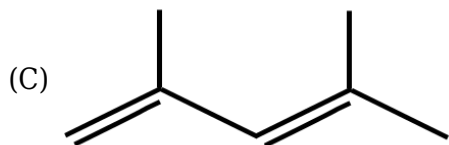
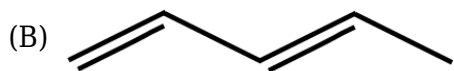
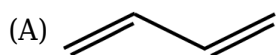
(A) Formaldehyde + $\text{NH}_3 \rightarrow$ Urotropine

(B) Formaldehyde + $\text{KMnO}_4 \xrightarrow{\text{H}^\oplus}$ formic acid

(C) Acetone $\xrightarrow[\Delta]{\text{Conc. H}_2\text{SO}_4}$ phorone

(D) $\text{Me} - \overset{\text{O}}{\parallel} \text{C} - \text{Et} \xrightarrow{\text{mCPBA}} \text{Et} - \overset{\text{O}}{\parallel} \text{C} - \text{OMe}$

4) Which of the following diene will give different major product with 1 eq. HBr at -40°C as well as $+40^\circ\text{C}$.

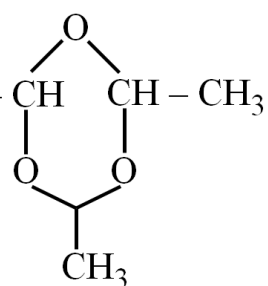


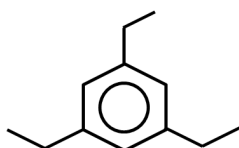
(D) All of these

SECTION-I (ii)

1) Option(s) having correctly matched major product of given reaction is /are:

(A) $\text{Me}_3\text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{OH} + \text{EtOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4} \text{Me}_3\text{C} - \overset{\text{O}}{\parallel} \text{C} - \text{OEt}$

(B) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow{\text{Conc. H}_2\text{SO}_4} \text{CH}_3 - \text{CH} \begin{array}{c} \diagup \text{O} \diagdown \\ \diagdown \text{O} \diagup \end{array} \text{CH} - \text{CH}_3$


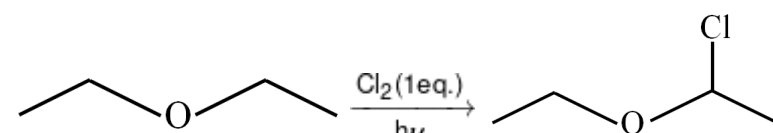
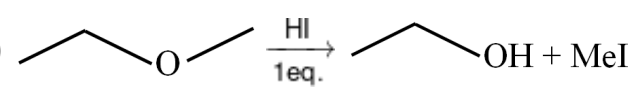
(C) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{CH}_3 \xrightarrow[\Delta]{\text{Conc. H}_2\text{SO}_4}$ 

(D) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3 \xrightarrow{\text{Dry HCl gas}} \text{CH}_3 - \text{C}(\text{CH}_3) = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{CH} = \text{C}(\text{CH}_3)_2$

2) Select incorrect statement(s)?

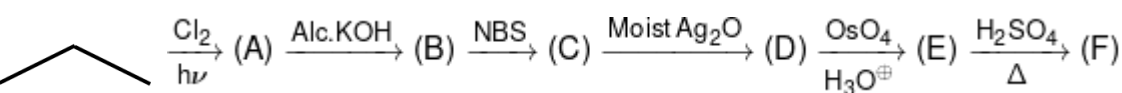
- (A) If sample of ether is giving red colour with $\text{FeSO}_4/\text{KCNS}$, it contains impurity of aldehyde.
- (B) All the alcohols giving red colour during Victor Meyer's test will give turbidity with Lucas reagent only after 30 minutes.
- (C) PhCHO and $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Me}$ can be differentiated by Fehling test.
- (D) Schiff's reagent is para rosaniline hydrochloride.

3) Select reaction(s) with correct major product

- (A) 
- (B) 
- (C) $\text{Ph-O-Me} \xrightarrow[\text{(excess)}]{\text{HI}} \text{PhOH} + \text{MeI}$
- (D) $\text{MeOMe} \xrightarrow[\Delta, \text{high P}]{\text{CO, BF}_3} \text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O-Me}$

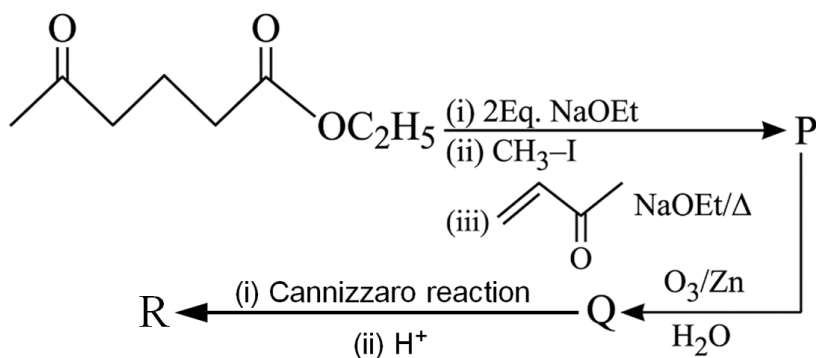
4) Select compound(s) giving positive Iodoform test.

- (A) $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Me}$
- (B) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{Ph}$
- (C) $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$
- (D) $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$

5) 

Select correct statement(s) about entire sequence.

- (A) It involves free radical substitution reaction.
- (B) It involves $\text{S}_{\text{N}}1$ reaction.
- (C) It involves pinacol pinacolone type rearrangement reaction
- (D) F can be used as tear gas.



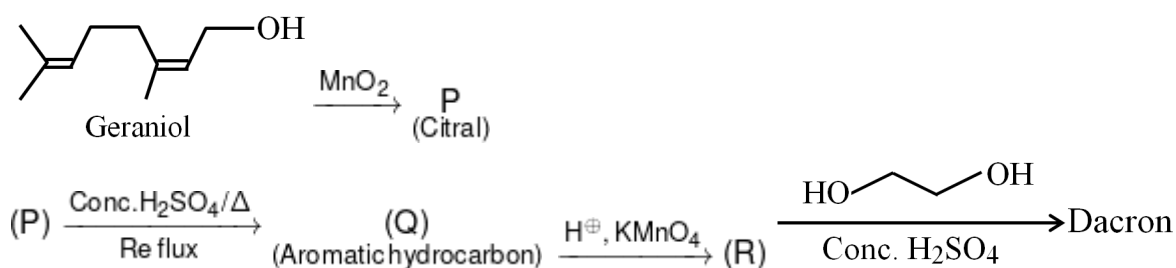
6)

Which of the following statement about product is correct?

- (A) D.U. (Degree of unsaturation) in product P is 5
- (B) Product R produces effervescence on addition of NaHCO_3
- (C) R gives positive iodoform test
- (D) Product R gives orange red colouration with 2,4-DNP

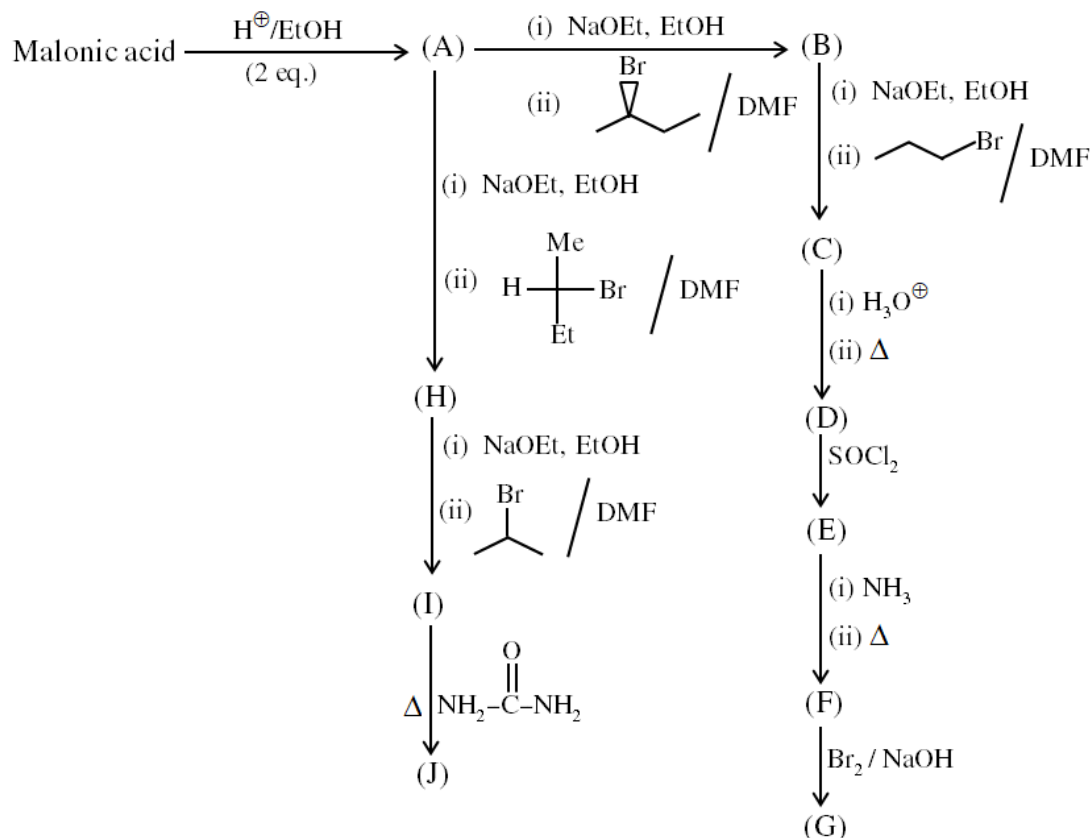
SECTION-II (i)

Common Content for Question No. 1 to 2



- 1) Calculate the total number of hyperconjugable $\alpha\text{-H}$ atoms in (Q)?
- 2) Calculate the degree of unsaturation of the major organic product (R)

Common Content for Question No. 3 to 4



3) Calculate total number of fractions obtained if all organic products formed in reaction from F to G undergo fractional distillation.

4) Calculate total number of chiral centers in product (J).

Common Content for Question No. 5 to 6

Isobutylene on reductive ozonolysis gives two organic compounds (A) & (B). A gives positive Fehling test. B on reaction with dilute NaOH followed by heating gives (C), which on reaction with CH_3MgBr followed by aqueous workup gives (D)

5) Starting with 1 mole isobutylene, calculate the mass (in grams) of major organic product (C) obtained, assuming 100% yield.

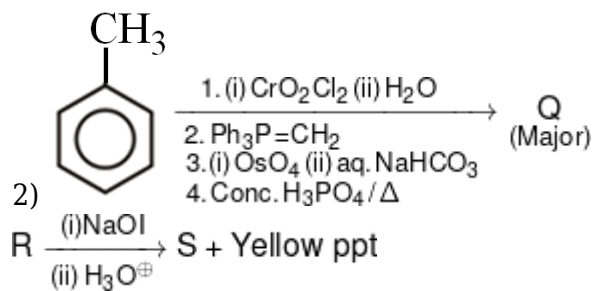
6)

Calculate the sum of locants of methyl substituents in major organic product (D) according to IUPAC rules

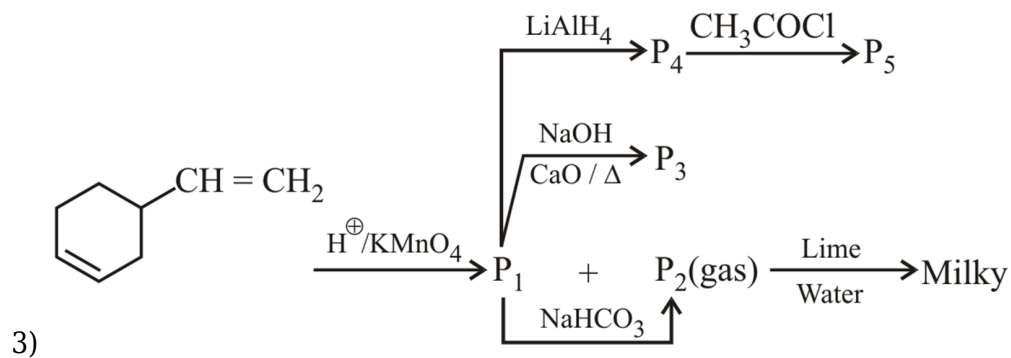
SECTION-II (ii)

1)

A polyhydric alcohol is acylated by using excess ethanoyl chloride in presence of alkali. It was observed that molecular mass of alcohol increased to 237% on acylation. If the molecular mass of polyhydric alcohol is 92, then the number of $-\text{OH}$ groups present in polyhydric alcohol are



R is benzenoid functional isomer of Q. What is the molecular mass of S?



(a) The number of carbon atoms present in the principal carbon chain of product P_3 obtained when P_1 is heated with sodalime.

(b) How many moles of gas P_2 released when one mole of P_1 is treated with sufficient amount of NaHCO_3 ?

(c) How many moles of CH_3COCl reacts with 1 mole of P_4 ?

(d) What is D.B.E. (Double bond equivalent) for P_1 ?

What will be the sum of answers of parts (a), (b), (c), (d).

ANSWER KEYS

PART-1 : MATHEMATICS

SECTION-I (i)

Q.	1	2	3	4
A.	B	B	A	C

SECTION-I (ii)

Q.	5	6	7	8	9	10
A.	A,B,C	A,B,D	A,C	A,D	A,C	A,B,D

SECTION-II (i)

Q.	11	12	13	14	15	16
A.	10.00	34.00	7.00	112.00	8.00	169.00

SECTION-II (ii)

Q.	17	18	19
A.	5	1	6

PART-2 : PHYSICS

SECTION-I (i)

Q.	20	21	22	23
A.	C	A	B	B

SECTION-I (ii)

Q.	24	25	26	27	28	29
A.	B,C	A,C	B,C	A,B,C	A,B,C,D	A,B,C,D

SECTION-II (i)

Q.	30	31	32	33	34	35
A.	1.33	0.73	6.00	6.43 to 6.45	2.67	2.59 to 2.60

SECTION-II (ii)

Q.	36	37	38
A.	9	34	6

PART-3 : CHEMISTRY

SECTION-I (i)

Q.	39	40	41	42
A.	C	C	A	A

SECTION-I (ii)

Q.	43	44	45	46	47	48
A.	B,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A,B,D

SECTION-II (i)

Q.	49	50	51	52	53	54
A.	4.00	6.00	2.00	2.00	49.00	6.00

SECTION-II (ii)

Q.	55	56	57
A.	3	122	13

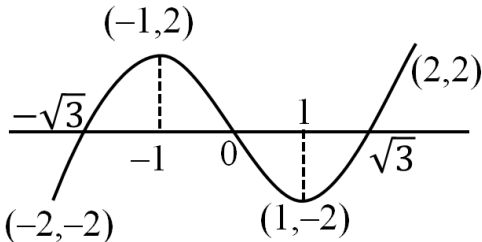
SOLUTIONS

PART-1 : MATHEMATICS

1)

$$Q - Q' = P \Rightarrow (Q \cdot e^{-x})' = -P e^{-x}$$

$$P \geq 0 \Rightarrow Q' \leq 0 \Rightarrow Q \text{ has no real roots}$$



2)

Let α, β, γ are roots of f

$f(x) = \alpha, f(x) = \beta, f(x) = \gamma$ have 3 distinct real roots

$$g(x) = x^3 - 3x = g(x) \in [-2, 2]$$

f has 3 real distinct real roots

$$\Rightarrow c \in (-2, 2)$$

$c = \pm 1$ are not possible

3)

Let $g(x) = \ln f(x) = x \ln x - (1-x) \ln(1-x)$. Differentiating with respect to x :

$$g'(x) = \ln x + 1 + \ln(1-x) + 1 = \ln(x(1-x)) + 2$$

Set $g'(x) = 0$ to find critical points :

$$\ln(x(1-x)) = -2 \Rightarrow x(1-x) = e^{-2}$$

There are two roots $\alpha, \beta \in (0, 1)$ corresponding to a local maximum and local minimum.

The product of the extrema is :

$$M \cdot m = f(\alpha) f(\beta) = \alpha e^{2(1-\alpha)} \cdot \beta e^{2(1-\beta)} = \alpha \beta e^{4-2(\alpha+\beta)}$$

$$M \cdot m = e^{-2} \cdot e^{4-2(1)} = e^{-2} \cdot e^2 = 1$$

4)

A tangent to $C : y = \frac{t^4}{4} + \frac{t^2}{2}$ at $x = t$ has the equation :

$$y = (t^3 + t)x - \left(\frac{3t^4}{4} + \frac{t^2}{2} \right)$$

Comparing with $y = mx - c(m)$, we get $m = t^3 + t$ and $c(m) = \frac{3t^4}{4} + \frac{t^2}{2}$.

$$h(m) = H(t) = \frac{3t^4}{4} + \frac{t^2}{2} - 2(t^3 + t)$$

$$H'(t) = 3t^3 + t - 2(3t^2 + 1) = t(3t^2 + 1) - 2(3t^2 + 1) = (t - 2)(3t^2 + 1)$$

Setting $H'(t) = 0$ gives a unique real root $t = 2$. The corresponding value m_0 is :

$$m_0 = m(2) = 2^3 + 2 = 10$$

Correct Option: (C)

$$5) \quad g' = \frac{xf' - f}{x^2} \text{ and } (xf' - f)' = xf'' < 0 \Rightarrow g(x) < 0$$

$$h' = \frac{(1-x)f' - f}{(1-x)^2} \text{ and } ((1-x)f' - f)' = (1-x)f'' < 0 \Rightarrow h'(x) > 0$$

6)

A tangent to $y = f(x)$ at $x = t$ has the equation $y - f(t) = f'(t)(x - t)$. For it to pass through $(0,0)$, it must satisfy $-f(t) = -t f'(t) = 0$.

(A) $f(t) = e^t : e^t - te^t = 0 \Rightarrow t = 1$. Exactly 1 tangent.

(B) $f(t) = t^4 - 2t^2 + 2 : t^4 - 2t^2 + 2 - t(4t^3 - 4t) = 0 \Rightarrow -3t^4 + 2t^2 + 2 = 0 \Rightarrow 3t^4 - 2t^2 - 2 = 0$. This

quadratic in t^2 has exactly one positive root $\left(t^2 = \frac{1 + \sqrt{7}}{3}\right)$, leading to two real values for t . Exactly 2 tangents.

(C) $f(t) = 12t^2 - t^4 : 12t^2 - t^4 - t(24t - 4t^3) = 0 \Rightarrow 3t^4 - 12t^2 = 0 \Rightarrow 3t^2(t^2 - 4) = 0$

Roots are $t = 0, 2, -2$. Exactly 3 tangents.

(D) $f(t) = t \ln t : t \ln t - t(1 + \ln t) = -t = 0$. Since $t \in (0, \infty)$, there is no solution.

Exactly 0 tangents.

$$7) \quad f(x) = \frac{x^2 + 2}{\sqrt{x^2 + 1}} = \sqrt{x^2 + 1} + \frac{1}{\sqrt{x^2 + 1}} \text{ and } A(t) = 4$$

8)

$$f_k'(x) = x(4x^2 + 3kx + 2) \text{ and } \Delta = 9k^2 - 32$$

9)

Differentiating an integral equation via Leibniz Rule strips it down to a familiar second-order differential equation.

$$\text{Given: } f(x) + \int_0^x (t-x)f(t)dt = e^{-x}$$

Differentiate both sides with respect to x :

$$f(x) - \int_0^x f(t)dt = -e^{-x}$$

Differentiate a second time :

$$f'(x) - f(x) = e^{-x}$$

$$f''(x) - f'(x) + f'(x) - f(x) = e^{-x}$$

$$\Rightarrow \frac{d}{dx} (e^x (f'(x) - f(x))) = 1$$

$$\begin{aligned} \Rightarrow e^x (f'(x) - f(x)) &= x + c \\ x = 0 \Rightarrow c &= -2 \\ \Rightarrow f'(x) - f(x) &= (x - 2)e^{-x} \\ \Rightarrow e^{-x}f(x) &= -\frac{x}{2}e^{-2x} - \frac{e^{-2x}}{4} + e^{-2x} + d \\ x = 0 \Rightarrow d &= \frac{1}{4} \end{aligned}$$

10)

Employing $g_n(x) = P_n(x)e^{-x}$ cleverly wraps the derivative of a truncated Taylor series into a simple decreasing exponential function.

Let $g_n(x) = P_n(x)e^{-x}$.

$$g'_n(x) = (P_{n-1}(x) - P_n(x))e^{-x} = \frac{x^n}{n!}e^{-x}$$

- For even n , $g'_n(x) \leq 0$ everywhere, making $g_n(x)$ strictly decreasing on $\mathbb{R}$. Since $g_n(x) \rightarrow 0$ as $x \rightarrow \infty$ and $g_n(x) \rightarrow \infty$ as $x \rightarrow -\infty$, $g_n(x) > 0$ always. Consequently, $P_n(x) > 0$ for all x , so $P_n(x) = 0$ has NO real roots. (A) is CORRECT.

- For odd n , $g'_n(x)$ changes sign precisely at $x = 0$ (from positive to negative). It rises from $-\infty$, hits a single local maximum at $x = 0$, and decays to 0. Thus, $g_n(x)$ crosses the x -axis exactly once, giving $P_n(x)$ exactly one real root. (B) is CORRECT.

- For even n , $g_n(x)$ is strictly decreasing and has NO local extremum. (C) is incorrect.

- If $P_n(r) = 0$ and $P_{n-1}(r) = 0$, then $P_n(r) = P_{n-1}(r) + \frac{r^n}{n!} = \frac{r^n}{n!} = 0 \Rightarrow r = 0$.

However, $P_n(0) = 1 \neq 0$. They share no common roots. (D) is CORRECT.

Correct Options: (A), (B), (D)

11)

Let tangent at (x_n, y_n) intersect again at (x_{n+1}, y_{n+1})

$$\square x^3 - 6x^2 + 5x + 1 = mx + c \text{ has roots } x_n, x_n, x_{n+1}$$

$$\Rightarrow 2x_n + x_{n+1} = 6, x_0 = 1$$

12)

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$$\square x^3 - 6x^2 + 5x + 1 = mx + c \text{ has roots } x_n, x_n, x_{n+1}$$

$$\Rightarrow 2x_n + x_{n+1} = 6, x_0 = 1$$

13)

To minimize any absolute deviation function $f(x) = \sum w_i |x - a_i|$, you just find the weighted median where the cumulative weight first crosses the 50% mark.

The total weight of the coefficients here is $1 + 2 + \dots + 10 = 55$, making our exact tipping point 27.5

Counting up from the left, the cumulative sum hits 21 at $x = 6$ and crosses the threshold to reach 28 at $x = 7$, locking $x = 7$ as the absolute minimum.

14)

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15)

Differentiating the clearance function $L(\theta) = \frac{27}{\sin \theta} + \frac{8}{\cos \theta}$ and setting it to zero isolates the critical angle at $\tan^3 \theta_c = \frac{27}{8}$, immediately solving the first part as $27 \cot^3 \theta_c = 8$.

Taking the cube root yields $\tan \theta_c = \frac{3}{2}$, which forms a standard $2-3-\sqrt{13}$ right triangle where $\theta_c = \frac{3}{\sqrt{13}}$ and $\cos \theta_c = \frac{2}{\sqrt{13}}$.

Substituting these ratios back into the original $L(\theta)$ equation gives the maximum gives the maximum pipe length

$$L_{\max} = 13\sqrt{13}. \text{ meaning } \frac{L_{\max}^2}{13} = 169.$$

16)

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$$L_{\max} = 13\sqrt{13}. \text{ meaning } \frac{L_{\max}^2}{13} = 169.$$

17) Tangency between dual parameter curves reduces to a system linking the equations and their derivatives.

If $y = c^x$ and $y = x^5$ touch, they share the same function value and derivative at some coordinate x .

$$1. c^x = x^5$$

$$2. c^x \ln c = 5x^4$$

Divide equation (2) by equation (1) :

$$\frac{c^x \ln c}{c^x} = \frac{5x^4}{x^5} \Rightarrow \ln c \frac{5}{x} \Rightarrow x = \frac{5}{\ln c}$$

Substitute x back into equation (1), noting that $c^x = e^{x \ln c}$:

$$e^{\left(\frac{5}{\ln c}\right) \ln c} = \left(\frac{5}{\ln c}\right)^5 \Rightarrow e^5 = \left(\frac{5}{\ln c}\right)^5$$

Taking the fifth root of both sides :

$$e = \frac{5}{\ln c} \Rightarrow 5 \ln c = \frac{25}{e}$$

18) By operating wholly in the logarithmic domain, a fearsome-looking minimum neatly collapses to $1 - 1/e$.

To minimize $f(x) = \frac{x^x e^x}{(x+1)^{x+1}}$, consider $g(x) = \ln f(x)$.

$$g(x) = x \ln x + x - (x+1) \ln(x+1)$$

Differentiate $g(x)$:

$$g'(x) = (\ln x + 1) + 1 - (\ln(x+1) + 1) = \ln x - \ln(x+1) + 1 = \ln \left(\frac{ex}{x+1} \right)$$

Set $g'(x) = 0$ to find the critical point :

$$\frac{ex}{x+1} = 1 \Rightarrow ex = x+1 \Rightarrow x(e-1) = 1 \Rightarrow x = \frac{1}{e-1}$$

Now, evaluate the original function f at this minimum point $x^0 = \frac{1}{e-1}$. Note that $x^0 + 1 =$

$$\frac{1}{e-1} + 1 = \frac{e}{e-1}$$

$$M = f(x_0) = \frac{(1/(e-1))^{1/(e-1)} e^{1/(e-1)}}{(e/(e-1))^{e/(e-1)}}$$

Group terms intelligently :

$$M = \frac{e^{1/(e-1)} \cdot (e-1)^{1/(e-1)}}{e^{e/(e-1)} \cdot (e-1)^{e/(e-1)}}$$

$$M = e^{\frac{1}{e-1} - \frac{e}{e-1}} \cdot (e-1)^{\frac{1}{e-1} - \frac{e}{e-1}}$$

$$M = e^{-1} \cdot (e-1)^1 = \frac{e-1}{e} = 1 - \frac{1}{e}$$

Calculating the value: $1 - \frac{1}{2.71828} \approx 1 - 0.3678 = 0.6321$. The quantity $100 \times M = 63.21$
Final Answer: 63.21

19) Converting the degree-4 polynomial inequality via the reciprocal substitution $t = \frac{1}{x}$ + brilliantly simplifies the optimization

Divide the inequality $x^4 - 4x^3 + kx^2 - 4x + 1 \geq 0$ by $x^2 > 0$ (if $x = 0$, $1 \geq 0$ obviously holds):

$$x^2 + \frac{1}{x^2} - 4 \left(x + \frac{1}{x} \right) + k \geq 0$$

Let $t = x + \frac{1}{x}$. We know that $|t| \geq 2$. Substitute $x^2 + \frac{1}{x^2} = t^2 - 2$:

$$t^2 - 2 - 4t + k \geq 0 \Rightarrow t^2 - 4t + k - 2 \geq 0$$

We need to find the absolute minimum of the quadratic $g(t) = t^2 - 4t - 2$ on the

domain $(-\infty, -2] \cup [2, \infty)$. The unconstrained vertex of this parabola is at $t = \frac{-4}{2} = -2$. Thus, for the restricted domain, the minimum occurs exactly at the boundary $t = 2$. Minimum value

is $g(2) = 2^2 - 4(2) - 2 = -6$. For the inequality to hold, we require $-6 + k \geq 0 \Rightarrow k \geq 6$. The smallest value of k is 6.

PART-2 : PHYSICS

20)

Correct Answer is (C)

21)

Correct Answer is (A)

22)

Correct Answer is (B)

23)

Correct Answer is (B)

24)

Correct Answer is (B,C)

25)

Correct Answer is (A,C)

26)

Correct Answer is (B,C)

27)

Correct Answer is (A,B,C)

28)

Correct Answer is (A,B,C,D)

29)

Correct Answer is (A,B,C,D)

30)

Correct Answer is (1.33)

31)

Correct Answer is (0.73)

32)

Correct Answer is (6.00)

33)

Correct Answer is (6.43 to 6.45)

34)

Correct Answer is (2.67)

35)

Correct Answer is (2.59 to 2.60)

36)

Correct Answer is (9)

37)

Correct Answer is (34)

38)

Correct Answer is (6)

PART-3 : CHEMISTRY

39) F is isobutyl chloride

40)

The correct answer is option (C)

41)

The correct answer is option (A)

42) The correct answer is option (A)

43) (A) Ether will be major product

(B) Paraldehyde

(C) Hexamethylbenzene

(D) Phorone

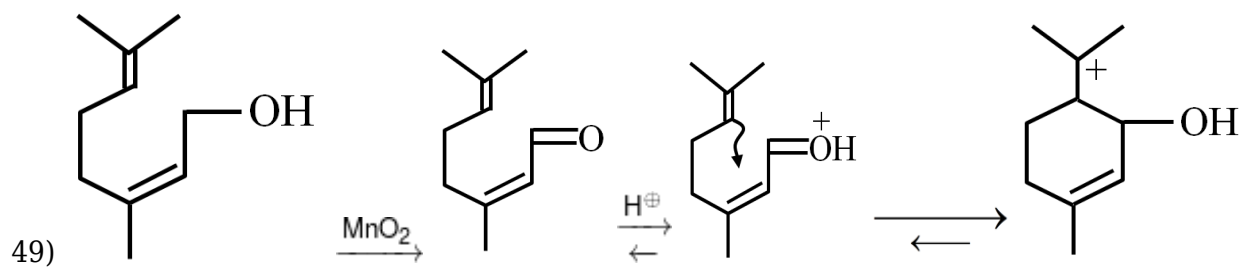
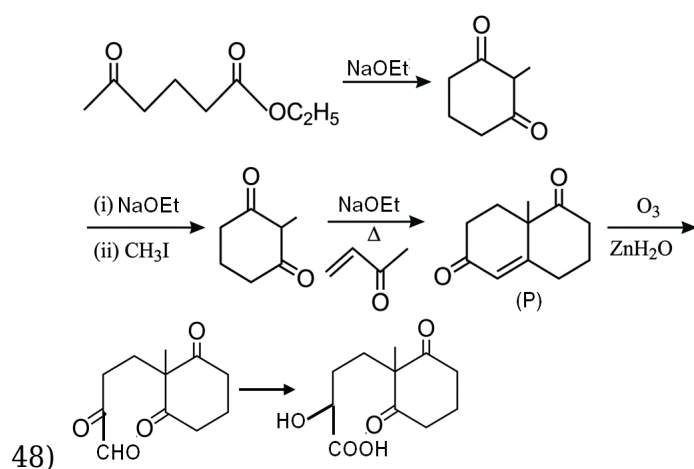
44) (A) It indicates hydroperoxides

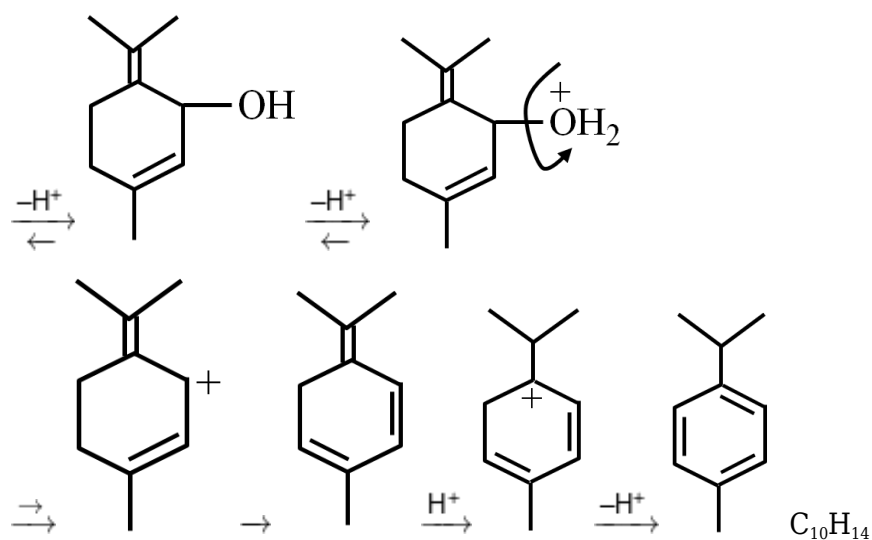
(B) PhCH_2OH gives immediate turbidity

45) The correct answer is option (A,B,C,D)

46) The correct answer is option (A,B,C,D)

47) The correct answer is option (A,B,C,D)





50)

Terephthalic acid

51) The correct answer is (2)

52) Correct answer is (2)
[Barbituric acid derivative]

53) Mesityl oxide (0.5 mole)

54) $Me_2C(OH)C=Me_2$ ($2 + 4 = 6$)

55) $ROH \xrightarrow{CH_3COCl} ROCOCH_3$

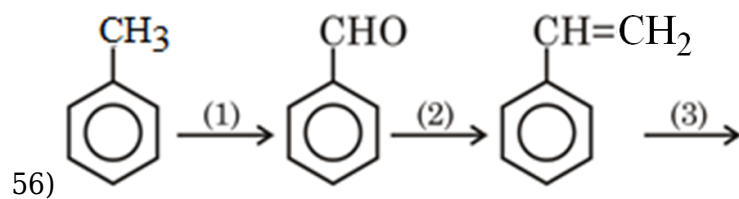
Here H is replaced by $COCH_3$ i.e. one H of mass 1 is replaced by one $COCH_3$ of mass 43.

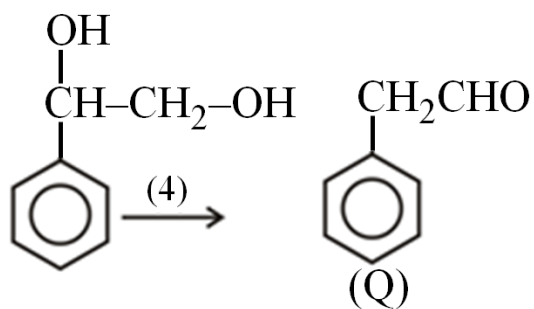
□ Mass increased is 42.

□ Mass of acylated product is = $92 \times \frac{237}{100} = 218$

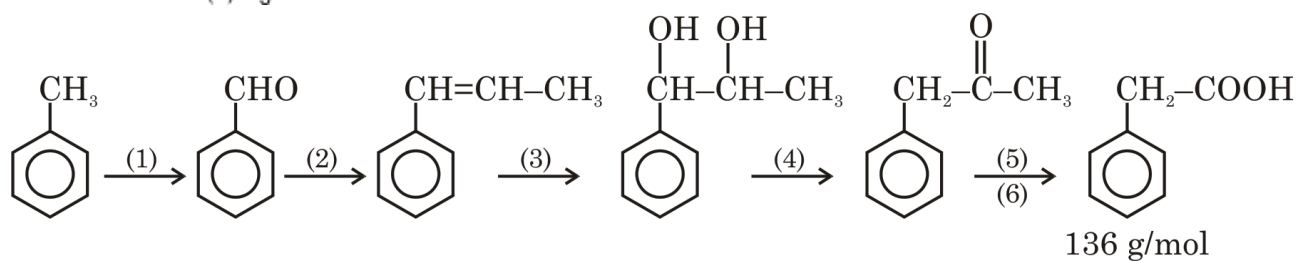
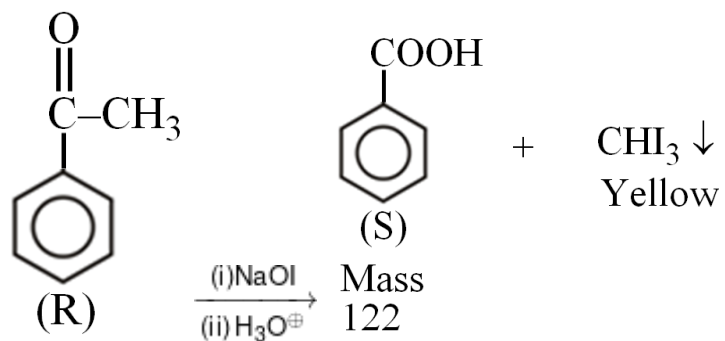
□ increase in mass = $218 - 92 = 126$

□ number of OH = $126/42 = 3$





Functional isomer of Q



57)

$$a = 4$$

$$b = 3$$

$$c = 3$$

$$d = 3$$

Sum of answer (a + b + c + d)

$$4 + 3 + 3 + 3 = 13$$